Research Problem Statement and Proposed Solution (System-Prompted Context)

Large language models (LLMs) frequently operate under system-level instructions that define their role, expertise, or domain specialization (e.g., “You are a helpful assistant specialized in Artificial Intelligence”). While such system prompting improves task alignment and domain-focused reasoning, it can also introduce unintended response bias. When the model is strongly conditioned to assume domain expertise, it may attempt to interpret every incoming query through the same domain lens, even when sufficient supporting evidence is unavailable. This behavior increases the likelihood of **domain-consistent hallucinations**, where the generated answer appears contextually appropriate but is factually incorrect.

This phenomenon can be empirically observed in stateless LLM deployments (e.g., API-based models without conversation memory). For example, when an LLM such as 4o-mini is given the system instruction *“You are a helpful assistant specialized in AI”* and is then asked the question *“Who is Prof. Gulzar Mehdi?”*, the model may generate a response claiming that the individual is an AI researcher or has conducted research in artificial intelligence, even when no verified evidence supports this claim. In this case, the hallucination is not triggered by retrieval failure or conversational context but by **prompt-induced domain bias**, where the model attempts to align its output with the imposed specialization. This demonstrates that system prompting alone can influence the model’s internal probability distribution, encouraging domain-aligned but fabricated information generation when factual grounding is absent.

These observations highlight a critical limitation in current LLM deployment practices: **system-prompt conditioning improves task focus but does not ensure factual reliability**. To address this challenge, we propose a multi-stage verification architecture that integrates an independent **Critic-LLM (C-LLM)** validation layer. The C-LLM operates as a post-generation evaluator that checks factual consistency, verifies claims against available evidence or knowledge sources, and flags potentially fabricated statements before the final response is delivered to the user. By separating generation and verification into distinct reasoning stages, the proposed framework reduces prompt-induced hallucination risk and significantly improves the trustworthiness and reliability of system-prompt-conditioned LLM applications.